

You are a senior full-stack developer. Build a production-ready "Team Task Manager" web application using the MERN stack:

Tech Stack:

- Frontend: React.js + Tailwind CSS
- Backend: Node.js + Express.js
- Database: MongoDB (Mongoose ODM)
- Authentication: JWT-based authentication

Core Features:

1. Authentication & Authorization:

- User signup and login
- Password hashing using bcrypt
- JWT token generation and verification
- Role-based access control (Admin, Member)

2. User Roles:

- Admin:
 - Create and manage projects
 - Add/remove team members
 - Assign tasks
- Member:
 - View assigned projects and tasks
 - Update task status

3. Project Management:

- Create, update, delete projects
- Add members to projects
- Each project contains multiple tasks

4. Task Management:

- Create tasks with:
 - Title
 - Description
 - Assigned user
 - Status (To Do, In Progress, Completed)
 - Due date
- Update and delete tasks
- Filter tasks by status, user, and due date

5. Dashboard:

- Show:
 - Total tasks
 - Completed tasks
 - Pending tasks
 - Overdue tasks
- Display tasks assigned to the logged-in user

Backend Requirements:

- RESTful API structure
- Proper folder structure (controllers, models, routes, middleware)
- Mongoose schemas with relationships:
 - User
 - Project
 - Task
- Validation using middleware or libraries (Joi or express-validator)
- Error handling middleware
- Role-based route protection middleware

Frontend Requirements:

- Clean UI using Tailwind CSS
- Pages:

- Login / Signup
- Dashboard
- Project list
- Task management view
- Use React Router for navigation
- State management using Context API or Redux Toolkit
- API integration using Axios
- Protected routes based on authentication

Database Design:

- User:
 - name, email, password, role
- Project:
 - title, description, members (array of user IDs), createdBy
- Task:
 - title, description, projectId, assignedTo, status, dueDate

Additional Requirements:

- Form validation on frontend and backend
- Use environment variables for sensitive data
- Modular, scalable code structure